

SEED STENCILS: HIGH-ORDER FINITE DIFFERENCES AND RBF-FD THROUGH MATERIAL EDGES TOO STEEP FOR THE GRID

BRADLEY P. MARTIN

ABSTRACT. Between a material whose parameters vary smoothly on the scale of the grid and one with a jump lies the case of an edge that is smooth but changes over a distance δ far smaller than the grid spacing h . Standard high-order finite differences sample the coefficients at the nodes and lose their order there until $h \lesssim \delta$, and the interface-aware stencils built for a jump do not apply, since there is no jump. We build stencils on the same equispaced (1-D) or scattered (2-D) nodes whose basis functions are not monomials but *seeds*: the $t = 0$ profiles of solutions of the wave equation that are polynomial in time, continued through the edge by ordinary differential equations in the coordinate normal to it. The seeds form an extended complete Chebyshev system for every δ , so the stencil solve is nonsingular, and as $\delta \rightarrow 0$ they reduce to the translated polynomial basis of the interface-aware stencils, so one construction covers the jump, the sub-grid edge and the resolved edge. In one dimension, standard fourth-order finite differences have a knee at $h = \delta$ while the seed stencils are fourth order at every resolution with the errors of the jump case. In two dimensions, for the elastic wave equation discretised with radial basis function-generated finite differences (RBF-FD) on scattered nodes, the seeds of a straight edge reduce to triangular systems of ordinary differential equations in the normal coordinate; the seed-augmented stencils are stable at the standard hyperviscosity, cut the error through an unresolved edge by up to an order of magnitude or more on the node sets tested, and lead to the rule: seed when $\delta \leq h$. The same seeds marched along the true normal handle a curved edge at the same accuracy on those node sets.

1. INTRODUCTION

[Stub, #55: the sub-grid smooth edge between “smooth” and “jump”; what standard high-order FD does there (the knee at $h = \delta$); the thesis: one construction from the jump (algebra) through the ODE march to the resolved edge (Fornberg); contributions; relation to prior work from LITERATURE.md (#53); outline.]

1.1. Relation to prior work. *[Stub, #53 supplies this paragraph; #55 places it.]*

Date: September 20, 2026.

2. SETTING

[Stub, #55: the 1-D first-order system $\rho u_t = f_x$, $f_t = Ku_x$; the 2-D elastic system in five fields; equispaced and scattered node sets; RBF-FD with polynomial augmentation, hyperviscosity, RK4; the interface-aware stencils of the jump as background (translated basis, continuity matrices, the two straddling rows, curvature by local expansion); what the implementation leaves out (triple stencils, variable Lamé parameters, the FD/RBF hybrid), stated as scope.]

3. SEEDS IN ONE DIMENSION

3.1. Seeds as the profiles of solutions polynomial in time. [Stub, #56: $u_{tt} = L_u u$ with $L_u = \rho^{-1} \partial_x K \partial_x$; the chain $\phi_0 = 1$, $K\phi'_1 = K(x_e)$, $L_u \phi_k = k(k-1)c_e^2 \phi_{k-2}$ with $\phi_k(x_e) = \phi'_k(x_e) = 0$; the even/odd (u/f) alternation; first-order form so that K is never differentiated.]

3.2. Nested kernels and the jump as the limit. [Stub, #56: $\{1\} \subset \ker L_u \subset \{L_u g = \text{const}\} \subset \ker L_u^2 \subset \dots$; the seed weights converge to the interface-aware weights at first order in δ .]

3.3. Well-posedness. [Stub, #56: the seeds are an extended complete Chebyshev system (Pólya, Karlin–Studden), so the per-stencil solve is non-singular on any distinct nodes for every δ .]

3.4. Why the Taylor route does not reach a sub-grid edge. [Stub, #56: notes §1.5; the quasi-Trefftz comparison.]

3.5. Stencil assembly and cost. [Stub, #56: the Fornberg-style solve on the seeds; one ODE march per stencil.]

4. ONE-DIMENSIONAL RESULTS

[Stub, #57: test problem (medium, contrast, pulse, $\delta \in \{0, 0.0025, 0.01\}$); the Fourier pseudo-spectral reference; the knee of standard FD4 at $h = \delta$; seeds fourth order at every n with the jump case's errors; condition numbers 20–60 from $\delta = 10^{-5}$ to 1.]

5. SEEDS IN TWO DIMENSIONS

5.1. Design: a straight edge. [Stub, #58: the x' -polynomial ansatz turns the 2-D chain into triangular systems of 1-D ODEs in the normal coordinate; tangential dependence stays polynomial; tractions as flux variables.]

5.2. The elastic seeds. [Stub, #58: all 30 seeds of one stencil marched as one 600-state system anchored at the evaluation node; constant-material, jump-limit, residual and conditioning checks.]

5.3. Stencil assembly. [Stub, #58: 19-node seed rows for the elastic operator; Δ^3 rows on the naive 30-node footprint with the seeds annihilated; the seeded region (19δ) and its trim.]

5.4. Curved edges. *[Stub, #58: route (a), the seeds marched along the true normal through the foot point, zeroth order in curvature, the same approximation as the jump stencils; what route (b) would be.]*

6. TWO-DIMENSIONAL RESULTS

6.1. Test problems and references. *[Stub, #59: 1-D spectral at normal incidence; Fourier-in- x for oblique; product-grid for the curved $\delta > 0$; the 122,500-node jump-aware run for the curved $\delta = 0$.]*

6.2. Naive baseline. *[Stub, #59: the resolution floor hides the edge error in v ; the spurious u separates unresolved from resolved edges $10\times$ more sharply.]*

6.3. Stability. *[Stub, #59: spectra at $n = 900$ and 2500 , standard γ , every δ , flat and curved.]*

6.4. The flat sweep and the rule. *[Stub, #59: seeds fourth order through $\delta = 0.0025$ at every n , $20\times$ below naive in v and $67\times$ in spurious u at 19,600 nodes; crossover at $h \approx \delta$; a resolved edge 1.4 – $2\times$ worse seeded; the rule: seed when $\delta \leq h$.]*

6.5. Oblique incidence. *[Stub, #59: seeds 1.4 – $1.9\times$ over naive at 26.6° , the ablation worse than naive, every scheme about 2.5 from the converted- S floor, reported as it is.]*

6.6. The curved edge. *[Stub, #59: jump stencils at the resolution floor at every n ; seeds equal the flat seeds at their own floor, 2.2 – $7\times$ ($\delta = 0.005$) and 3 – $12\times$ ($\delta = 0.0025$) below naive; where route (a)'s geometry first shows; the fine-end rate and the trim; costs.]*

7. DISCUSSION

7.1. Limitations. *[Stub, #60: FD4 / degree 3 only, one contrast, at most 19,600 nodes, the flat translation symmetry unexploited, one march per stencil, the port's exclusions.]*

7.2. Future work. *[Stub, #60: curvilinear evaluation of the marched seeds; seeded-width tuning; the seeds' true Δ^3 as hyperviscosity right-hand sides; Mühlbach's recurrence in place of the dense solve; higher order; other contrasts; thin bands; variable Lamé parameters; 3-D; the oblique train on the curved geometry.]*

7.3. Conclusions. *[Stub, #60: the rule and the one-construction claim, in #53's wording.]*

ACKNOWLEDGMENTS

[Stub, #62: Bengt Fornberg, for the 2016-era work this construction extends (the dissertation and the interface treatment of [MF17a]); the AI-assistance statement crediting Claude (Fable 5.1) for the re-derivation, the code and the drafting.]

REFERENCES

- [AS55] D. N. de G. Allen and R. V. Southwell, *Relaxation methods applied to determine the motion, in two dimensions, of a viscous fluid past a fixed cylinder*, The Quarterly Journal of Mechanics and Applied Mathematics **8** (1955), no. 2, 129–145.
- [BCO94] I. Babuška, G. Caloz, and J. E. Osborn, *Special finite element methods for a class of second order elliptic problems with rough coefficients*, SIAM Journal on Numerical Analysis **31** (1994), no. 4, 945–981.
- [BGL17] Lehel Banjai, Emmanuil H. Georgoulis, and Oluwaseun Lijoka, *A Trefftz polynomial space-time discontinuous Galerkin method for the second order wave equation*, SIAM Journal on Numerical Analysis **55** (2017), no. 1, 63–86.
- [BO83] I. Babuška and J. E. Osborn, *Generalized finite element methods: their performance and their relation to mixed methods*, SIAM Journal on Numerical Analysis **20** (1983), no. 3, 510–536.
- [Cop71] W. A. Coppel, *Disconjugacy*, Lecture Notes in Mathematics, vol. 220, Springer, 1971.
- [FL11] Bengt Fornberg and Erik Lehto, *Stabilization of RBF-generated finite difference methods for convective PDEs*, Journal of Computational Physics **230** (2011), no. 6, 2270–2285.
- [For88] Bengt Fornberg, *Generation of finite difference formulas on arbitrarily spaced grids*, Mathematics of Computation **51** (1988), no. 184, 699–706.
- [HW97] Thomas Y. Hou and Xiao-Hui Wu, *A multiscale finite element method for elliptic problems in composite materials and porous media*, Journal of Computational Physics **134** (1997), no. 1, 169–189.
- [IGMS23] Lise-Marie Imbert-Gérard, Andrea Moiola, and Paul Stocker, *A space-time quasi-Trefftz DG method for the wave equation with piecewise-smooth coefficients*, Mathematics of Computation **92** (2023), no. 341, 1211–1249, arXiv:2011.04617.
- [IGMS24] ———, *Polynomial quasi-Trefftz DG for PDEs with smooth coefficients: elliptic problems*, 2024, arXiv:2408.00392.
- [Il’69] A. M. Il’in, *Differencing scheme for a differential equation with a small parameter affecting the highest derivative*, Mathematical Notes **6** (1969), 596–602.
- [KMPS16] Fritz Kretschmar, Andrea Moiola, Ilaria Perugia, and Sascha M. Schnepp, *A priori error analysis of space-time Trefftz discontinuous Galerkin methods for wave problems*, IMA Journal of Numerical Analysis **36** (2016), no. 4, 1599–1635.
- [KS66] Samuel Karlin and William J. Studden, *Tchebycheff systems: With applications in analysis and statistics*, Interscience, New York, 1966.
- [LL94] Randall J. LeVeque and Zhilin Li, *The immersed interface method for elliptic equations with discontinuous coefficients and singular sources*, SIAM Journal on Numerical Analysis **31** (1994), no. 4, 1019–1044.
- [M73] G. Mühlbach, *A recurrence formula for generalized divided differences and some applications*, Journal of Approximation Theory **9** (1973), no. 2, 165–172.

- [Mar16] Bradley P. Martin, *Application of RBF-FD to wave and heat transport problems in domains with interfaces*, Ph.D. thesis, University of Colorado Boulder, 2016, ProQuest 10151046.
- [MF17a] Bradley Martin and Bengt Fornberg, *Seismic modeling with radial basis function-generated finite differences (RBF-FD) – a simplified treatment of interfaces*, *Journal of Computational Physics* **335** (2017), 828–845.
- [MF17b] ———, *Using RBF-FD to solve heat transfer equilibrium problems in domains with interfaces*, *Engineering Analysis with Boundary Elements* **79** (2017), 38–48.
- [MFSC15] Bradley Martin, Bengt Fornberg, and Amik St-Cyr, *Seismic modeling with radial-basis-function-generated finite differences*, *Geophysics* **80** (2015), no. 4, T137–T146.
- [OZ07] Houman Owhadi and Lei Zhang, *Metric-based upscaling*, *Communications on Pure and Applied Mathematics* **60** (2007), no. 5, 675–723, arXiv:math/0505223.
- [OZ08] ———, *Numerical homogenization of the acoustic wave equations with a continuum of scales*, *Computer Methods in Applied Mechanics and Engineering* **198** (2008), no. 3–4, 397–406.
- [P22] G. Pólya, *On the mean-value theorem corresponding to a given linear homogeneous differential equation*, *Transactions of the American Mathematical Society* **24** (1922), no. 4, 312–324.
- [RST08] Hans-Görg Roos, Martin Stynes, and Lutz Tobiska, *Robust numerical methods for singularly perturbed differential equations*, 2nd ed., Springer Series in Computational Mathematics, vol. 24, Springer, 2008.
- [Sam01] A. A. Samarskii, *The theory of difference schemes*, Marcel Dekker, New York, 2001.
- [SG69] D. L. Scharfetter and H. K. Gummel, *Large-signal analysis of a silicon Read diode oscillator*, *IEEE Transactions on Electron Devices* **16** (1969), no. 1, 64–77.
- [SV67] M. H. Schultz and R. S. Varga, *L-splines*, *Numerische Mathematik* **10** (1967), no. 4, 345–369.
- [TS62] A. N. Tikhonov and A. A. Samarskii, *Homogeneous difference schemes*, *USSR Computational Mathematics and Mathematical Physics* **1** (1962), no. 1, 5–67.

Email address: 1bradley.martin1@gmail.com